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Students must be fully aware that any reasonable adjustments do not override academic standards and prescribed requirements of professional bodies.

VACOLIDAY DATABASE SYSTEMS REPORT

Tutorial group :5CS12
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Project Brief: Vacoliday

Vacoliday is a touristic company that offers high-end highly-customizable holidays, with a great emphasis on localness and well-being, in Vacomedita, a lovely town on the Mediterranean coast.

Vacoliday has several residences around the town of Vacomedita which they use to accommodate holiday makers. These residences offer two types of bookable accommodation: fully-integrated apartments and stand-alone rooms. The apartments are typically larger with one or more bedrooms and other rooms like kitchens, bathrooms, washing rooms, dining rooms, and lounges. The standalone rooms are essentially bedrooms with possibly an en-suite bathroom.

The company offers various attractive meal options that can also be booked. Collective meals can be taken at specific times at one of the stylish dining suites available at the Vacoliday residences. However, meals can also be delivered to people's accommodation, if pre-booked in advance.

Besides, Vacoliday offers a wide range of exciting bookable experiences. These experiences can take place on-site at a specific residence or off-site, out and about the area surrounding Vacomedita. There are three main kinds of on-site experiences: fitness classes (such as yoga, pilates or any other well-being physical activities) under the guidance of a fitness instructor; craft workshops during which Vacoliday customers can learn a local craft with a workshop instructor; and cooking classes for which holidaymakers are taught how to prepare local dishes by a chef. On the other hand, offsite experiences are led by a guide and come mainly under two broad categories: excursions to places of interest and walking hikes. Thus, hikes are on foot and involve walking in the countryside or wilderness around Vacomedita. Excursions are run by a driver on a bus and allow the visiting of places of interest, beauty spots and other landmarks across the stunning Vacomedita region.

In terms of staffing, Vacoliday relies on highly-trained employees with well-defined roles. Guides oversee all off-site experiences. Drivers use their buses to accompany people on excursions. Chefs take care of meals but also teach cooking classes. Housekeeping attendants are responsible for the good maintenance of their allocated accommodations. Finally, instructors come under two groups: the fitness instructors who run the exercise classes for well-being and the craft instructors who lead and deliver the workshops in which people learn exciting local craft skills.

QUESTION FOR VACOLIDAY & REFLECTIVE COMMENTARY

I have several questions I need answered to complete the database architecture.

Q1). Are there any seasonal experiences that would require a start and end month for the booking seasons?

Answer: All our experiences vary continuously and there is no seasonality. Storing the right info for every experience like name, start date time, end date time, a description, the cost, age requirements, accessibility requirements and the maximum capacity is essential.

Q2). Must the meals be booked in advance, or can they be booked during the trip?

Answer: Meals can be booked at any time, but it has to be at least one day before.

Q3). How many staff make up the various teams in your Business? I.e. the Chef's, Housekeepers, and Instructors?

Answer: Q3) No set numbers, the numbers are irrelevant. What matters is that the right staff is allocated to cover the right experiences, meals, and accommodations.

Q4). What information are we required to hold on the databases for, "the Holidaymakers, there Bookings, the Residences, the Staff, their Experiences and their Meals?

Answer: For every guest, storing their personal details, contact details, date of birth, card payment details, dietary requirements, fitness levels and possible disabilities or impairments is essential. For every booking, storing the booking reference number, the booking date and time and the cost is important. For every residence, storing the name, address and description of the residence would be helpful. For every staff, we need For every member of staff, storing their personal details, contact details, experience, expertise, and qualifications is required. For every experience, we need the name, start date time, end date time, a description, the cost, age requirements, accessibility requirements and the maximum capacity are needed. For every meal, the name, a description, a price and the date and time of the meal are needed.

Q5). What is the accessibility of each experience, are there some experiences that are not suitable to disabled individuals and therefore will need to be noted?

Answer: Every experience has a different fitness level appropriateness which varies from experience to experience. Therefore, storing the fitness level appropriateness for every experience is essential.



Thomas Mifsud
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Q1).

Are there any seasonal experiences requiring a start and end month for the booking seasons?

Q2).

Must the meals be booked in advance, or can they be booked during the trip?

Q3).

**How many staff make up the various teams in your Business? I.e.
the Chef's, Housekeepers, and Instructors**

Q4).

What information are we required to hold on the databases for, "the Holidaymakers, their Bookings, the Residences, the Staff, their Experiences and their Meals.

Q5).

What is the accessibility of each experience, are there some experiences that are not suitable for disabled individuals and therefore will need to be noted?

Proof of business questions Submission.

A brief rundown of the information obtained from my five business questions.

- The answer for Q4 provides much needed information on specific attributes that will need to be included in the EERD. I shall refer to this when completing my attributes dictionary.
- Q3 clarifies the issue I was having about specific staffing number and simplifies the issue immensely. I shall not have to worry about team sizes with this EERD as the client wishes to keep it simple.
- Q5 provides more clarity on necessary attributes as it identifies that both skill and disability levels will need to be included in the database system.
- Q1 clarifies that there are no seasonal experiences, rather experiences tend to vary throughout the year without any set season. It also provides some more attributes to be including in the attribute data dictionary.
- Finally, Q2 clarifies that the meals do not have to be booked in advance but must be booked 24 hours before the meal. As such, I shall have to redesign my EERD to accommodate this need.

ENTITIES

From Vacoliday's brief It is clear that there are 6 main Entities. I have provided these Entities along with a brief explanation in the entity data dictionary table below.

Entity Data Dictionary Table

<i>Entity name</i>	<i>Brief Explanation</i>
w1832388_Holidaymakers	The Client who books a residence
w1832388_Booking	The Clients booking information
w1832388_Accommodations	Hirable property's belonging to Vacoliday
w1832388_Residences	Where the accommodation Is located, and some Experiences take place. Also, the Residences Are the locations of the collective males. (Which will be discussed later in the report)
w1832388_Staff	Vacoliday Staff and workers.
w1832388_Experiences	Activities provided on and off site by Vacoliday
w1832388_Meals	Bookable eating options provided by Vacoliday

SPECIALISATIONS & GENERALISATION

Vacoliday's Database system will have to contain multiple specialised and generalised entities due to the wide variety of experiences and accommodation options they provide. I have identified what I believe to be all of the main specialised and generalised entities that will need to be outlined within the EERD. I have provided these entities within the specialised data dictionary table below, along with some brief explanations.

Specialization Data Dictionary Table

<i>General entity</i>	<i>Specialized entity</i>	<i>Brief explanation</i>
w1832388_Accommodation	w1832388_FullyIntegrated_apartments w1832388_standAloneRooms w1832388_With_ensuite w1832388_Without_ensuite	The brief describes two types of Accommodation, fully integrated apartments with kitchens and multiple bedrooms, and standalone rooms. The standalone room option also provides the option of an ensuite. Hence the specialization of the ensuite for the standalone room.
w1832388_Staff	w1832388_Drivers w1832388_Chefs w1832388_Housekeeping w1832388_Instructors w1832388_Guides	The brief outlines four staff roles. That of Drivers, Chefs, Housekeepers, Instructors, and Guides
w1832388_Experiences	w1832388_On-site w1832388_Fitness w1832388_Classes w1832388_Crafts w1832388_Chefs Course w1832388_Offsite w1832388_Excursions w1832388_Walking w1832388_Hikes	The brief Provides an overview of the various experiences available to Holidaymakers On or around Vacoliday's residences. It divides them into two specializations, "On-site and off-site", each with their own sub-specialisations. The three options for on-site experiences are fitness classes, crafts, and chef courses. Whereas offsite has two options of excursions and Walking Hikes.
w1832388_Meals	w1832388_Collective_Meals w1832388_Delivered_Meals	The brief explains that there are two main options provided for holidaymakers. they could either eat a collective meal with fellow holiday goers or have food delivered to their room.

RELATIONSHIPS & MULTIPLICITIES

Now that I have identified the component entity's, I can start determining the Relationship & Multiplicities for the EERD. Below I have provided a relationship and Multiplicities Data Dictionary Table. This table outlines the relationship of each entity to one another, It also provides the justifications for the entity's participation & cardinality.

Relationship & Multiplicities Data Dictionary Table

Entity name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity (4 statements for each relationship)
w1832388_Holidaymakers	0..1	Make	0..*	w1832388_Booking	Holidaymakers do not have to make a booking
					Holiday makers can make up to one booking.
					There can be zero bookings
					There can be many bookings
w1832388_Booking	0..* 0..*	Reserves Single Rooms in Reserves Apartments in	1..* 1..*	w1832388_Accommodation	A booking will have at least one Accommodation attached
					A booking may have many Accommodations attached
					An Accommodation may not have any bookings
					An Accommodation may have many bookings
w1832388_Accommodation	1..*	Is Located In	1..1	w1832388_Residences	Accommodation Is Located In At Least one Residence
					Accommodation Is Located In At most one Residence
					Residences Contain at least one Accommodation
					Residences can Contain many Accommodations
w1832388_Residences	1..* 1..* 1..* 1..* 1..*	Has chefs	0..*	w1832388_Staff	A residence may not have any members of staff
		Has housekeepers	0..*		A residence may have many members of staff
		Has Guides	0..*		Staff members will be assigned to at least one residence
		Has instructors	0..*		Staff members may be assigned to many residents
		Has Drivers	0..*		
w1832388_Staff	1..* 1..*	Guides Run Driver Transport to	0..* 0..*	w1832388_OffSite_Experiences	Off-Site Experiences may have only one Guides and one Drivers
					Off-Site Experiences may have many Guides and many Drivers
					Guides and Drivers may be assigned to no Off-Site Experiences

					Guides and Drivers may be assigned to many Off-Site Experiences
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Entity name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity (4 statements for each relationship)
w1832388_Holidaymakers	1..*	Books Collective Meal	0..*	w1832388_Meals	A Holidaymaker may not book any Meals
	1..*	Books Delivery Meal	0..*		A Holidaymaker may not book many Meals
					A Meals will have at least one Holidaymaker
					A Meals will have many Holidaymaker
w1832388_Collective_Meals	0..*	Are served in	1..*	w1832388_Residences	Collective Meals Are served in at Least one Residences
					Collective Meals can be served in many Residences
					Residences may have no Collective Meals
					Residences may have many Collective Meals
w1832388_Deliver_Meals	0..*	Delivered to	1..1	w1832388_Accommodation	Deliver Meals are delivered to At Least one Accommodation
					Deliver Meals are delivered to At most one Accommodation
					Accommodation may have no Deliver Meals
					Accommodation may have many Deliver Meals
w1832388_Holidaymakers	1..*	Book an on-site Experience	0..*	w1832388_Experiences	Holidaymakers may have no Experiences Booked
	1..*	Book an off-site Experience	0..*		Holidaymakers may have many Experiences Booked
					Experiences may have no Holidaymakers Booked
					Experiences may have many Holidaymakers Booked
w1832388_OnSite_Experiences	0..*	Takes place in	1..*	w1832388_Residences	On-Site Experiences Takes place in at Least one Residence
					On-Site Experiences Takes place in at most one Residence
					I Residence may have no On-Site Experiences
					I Residence may have many On-Site Experiences

Entity name	Multiplicity	Relationship	Multiplicity	Entity name	Brief justifications for the multiplicity (4 statements for each relationship)
w1832388_Staff	1..*	Instructor Run	0..*	w1832388_OnSite_Experiences	On-Site Experiences may have only one Instructor or one Chef
	1..*	Chef Runs	0..*		On-Site Experiences may have many Instructors or many Chefs
					Instructors and Chefs may be assigned to no On-Site Experiences
					Instructors and Chefs may be assigned to many On-Site Experiences
w1832388_Housekeepers	0..*	Cleans	1..*	w1832388_Accommodation	Accommodation may have No Housekeepers
					Accommodation may have many Housekeepers
					Housekeepers may be assigned to no Accommodation
					Housekeepers may be assigned to many Accommodation
w1832388_Housekeepers	0..*	Cleans	1..*	w1832388_Residences	Residences may have No Housekeepers
					Residences may have many Housekeepers
					Housekeepers may be assigned to no Residences
					Housekeepers may be assigned to many Residences
w1832388_Chefs	1..*	Chefs Prepare	0..*	w1832388_Meals	Meals may have only one Chef
					Meals may have many Chef
					Chef may be assigned to no Meals
					Chef may be assigned to many Meals

ATTRIBUTES & PRIMARY KEYS

Now that I have determined the relationships between the entities, it is possible to start identifying the attributes that belong to them. Thanks to the answers provided to my questions I was able to provide attributes for all of the outlined data entry. I also identified a number of attributes that I think would also be helpful to include. Below, I have provided an attribute data dictionary table, outlining the entities and their assorted attributes and primary keys with a brief explanation of each.

Attributes Data Dictionary Table

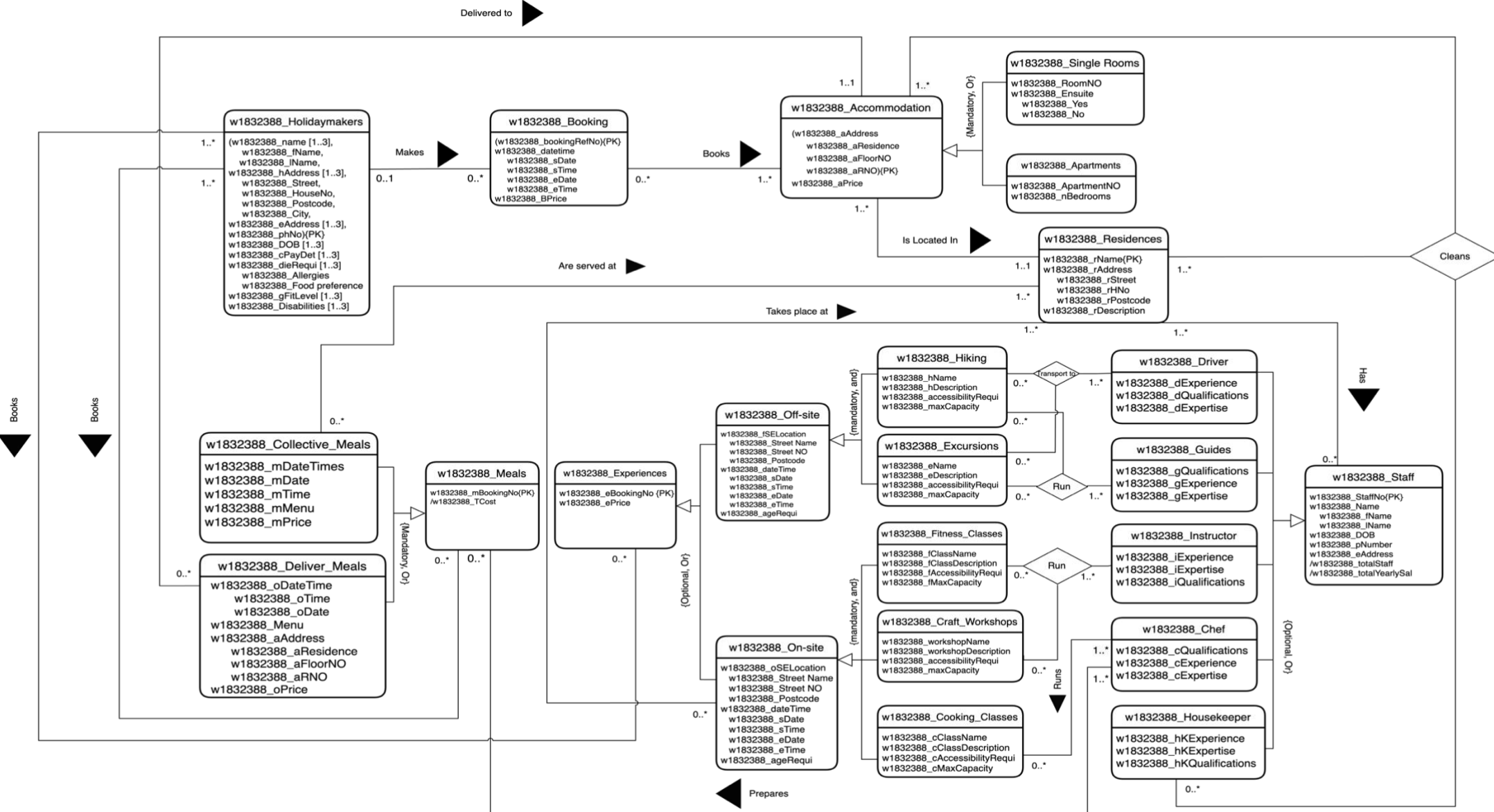
Entity name	Attributes for each entity (include PK)	Brief explanation
w1832388_Holidaymakers	(w1832388_name [1..3], w1832388_fName, w1832388_lName, w1832388_hAddress [1..3], w1832388_Street, w1832388_HouseNo, w1832388_Postcode, w1832388_City, w1832388_eAddress [1..3], w1832388_phNo){PK} w1832388_DOB [1..3] w1832388_cPayDet [1..3] w1832388_dieRequi [1..3] w1832388_Allergies w1832388_Food preference w1832388_gFitLevel [1..3] w1832388_Disabilities [1..3])	(Guest name, First name, Last name, Home address, Street name, House number, Postcode, City name, Email address, Phone number: Primary key) date of birth card payment details dietary requirements What are the guest allergies if any Is the guest, vegan, vegetarian etc.. fitness levels Does the guest have any disabilities
w1832388_Booking	(w1832388_bookingRefNo){PK} w1832388_datetime w1832388_sDate w1832388_sTime w1832388_eDate w1832388_eTime w1832388_BPrice	(Booking reference number: Primary key) Booking date and time Start date Start time End date End time Booking cost
w1832388_Residences	w1832388_rName{PK} w1832388_rAddress w1832388_rStreet w1832388_rHNo w1832388_rPostcode w1832388_rDescription	(Residence name: Primary key) Residence address Residence Street Name, Residence Street Number, Residence Postcode Description of Residence
w1832388_Staff	w1832388_StaffNo{PK} w1832388_Name w1832388_fName w1832388_lName w1832388_DOB w1832388_pNumber w1832388_eAddress /w1832388_totalStaff /w1832388_totalYearlySal	(Staff Number: Primary Key). personal details First Name. Last Name. Date of birth phone number. email address. Total number of employed staff Yearly Salary

Entity name	Attributes for each entity (include PK)	Brief explanation
w1832388_Driver	w1832388_dExperience w1832388_dQualifications w1832388_dExpertise	Driving experience Driving qualifications Driver expertise
w1832388_Chef	w1832388_cQualifications w1832388_cExperience w1832388_cExpertise	Chef qualifications Chef experience Chef expertise
w1832388_Housekeeper	w1832388_hKExperience w1832388_hKExpertise w1832388_hKQualifications	Housekeeping experience Housekeeping expertise Housekeeping qualifications
w1832388_Instructor	w1832388_iExperience w1832388_iExpertise w1832388_iQualifications	Instructor experience Instructor expertise Instructor qualifications
w1832388_Guide	w1832388_gQualifications w1832388_gExperience w1832388_gExpertise	Guide qualifications Guide experience Guide expertise
w1832388_Experiences	w1832388_eBookingNo {PK} w1832388_ePrice	(Experience Booking Number: Primary Key). Experiences Price.
w1832388_OnSite_Experiences	w1832388_oSELocation w1832388_Street Name w1832388_Street NO w1832388_Postcode w1832388_dateTime w1832388_sDate w1832388_sTime w1832388_eDate w1832388_eTime w1832388_ageRequi	Experience location location Street Name location Street Number location Postcode Date & Time Start date Start time End date End time Age requirements
w1832388_Fitness_Classes	w1832388_fClassName w1832388_fClassDescription w1832388_fAccessibilityRequi w1832388_fMaxCapacity	Fitness Classes Name Fitness Classes description Accessibility requirements Maximum capacity
w1832388_Craft_Workshops	w1832388_workshopName w1832388_workshopDescription w1832388_accessibilityRequi w1832388_maxCapacity	Workshops Name Workshops description Accessibility requirements Maximum capacity
w1832388_Cooking_Classes	w1832388_cClassName w1832388_cClassDescription w1832388_cAccessibilityRequi w1832388_cMaxCapacity	Cooking Classes Name Cooking Classes description Accessibility requirements Maximum capacity
w1832388_OffSite_Experiences	w1832388_fSELocation w1832388_StreetName w1832388_StreetNO w1832388_Postcode w1832388_dateTime w1832388_sDate w1832388_sTime w1832388_eDate w1832388_eTime w1832388_ageRequi	Experience location Off-site location Street Name Off-site location Street Number Off-site location Postcode Date & Time Start date Start time End date End time Age requirements

Entity name	Attributes for each entity (include PK)	Brief explanation
w1832388_Excursions	w1832388_eName w1832388_eDescription w1832388_accessibilityRequi w1832388_maxCapacity	Name of excursions description Accessibility requirements Maximum capacity
w1832388_Hiking	w1832388_hName w1832388_hDescription w1832388_accessibilityRequi w1832388_maxCapacity	Name of Hike Hike description accessibility requirements Maximum capacity
w1832388_Meals	w1832388_mBookingNo{PK} /w1832388_TCost	(Meal Booking number and: primary Key) Total Cost
w1832388_Collective_Meals	w1832388_mDateTimes w1832388_mDate w1832388_mTimes w1832388_mMenu w1832388_mprice	Meals, time, and date Meal's date Meal's time Current available dishes. Meal's price.
w1832388_Delivery_Meals	w1832388_oDateTime w1832388_oTime w1832388_oDate w1832388_Menu w1832388_aAddress w1832388_aResidence w1832388_aFloorNO w1832388_aRNO w1832388_oPrice	Time and date of order Times of order Date of order Current available dishes Residence In which the Accommodation is Located, Floor Number of Residence, Accommodation Room Number Order Price
w1832388_Accommodation	(w1832388_aAddress w1832388_aResidence w1832388_aFloorNO w1832388_aRNO){PK} w1832388_aPrice	(Residence In which the accommodation is located, Floor Number of Residence, Accommodation room number: Primary Key) Accommodation Price.
w1832388_Apartments	w1832388_ApartmentNO w1832388_nBedrooms	Apartment Number Number of bedrooms
w1832388_Single_Rooms	w1832388_RoomNO w1832388_Ensuite w1832388_Yes w1832388_No	Room Number Yes, it does contain an ensuite. No, it does not contain an ensuite.

CONCEPTUAL EERD

(see next page)



Logical ERD

Project Brief: Tourmato

Tourmato is a touristic company that offers exciting “off-the-beaten track” visiting tours around several cities across Europe. Essentially, Tourmato takes groups of customers around European cities to visit them and experience their unique atmospheres, under the direction of a local experienced tour guide. To provide a range of experiences, every city covered by Tourmato offers many visiting tours. A tour session is a visiting tour that has been assigned a specific start date and time, a start address, an end date and time, an end address, and a specific price. Tourmato customers can place a booking for a variety of exciting tour sessions, depending on what they like.

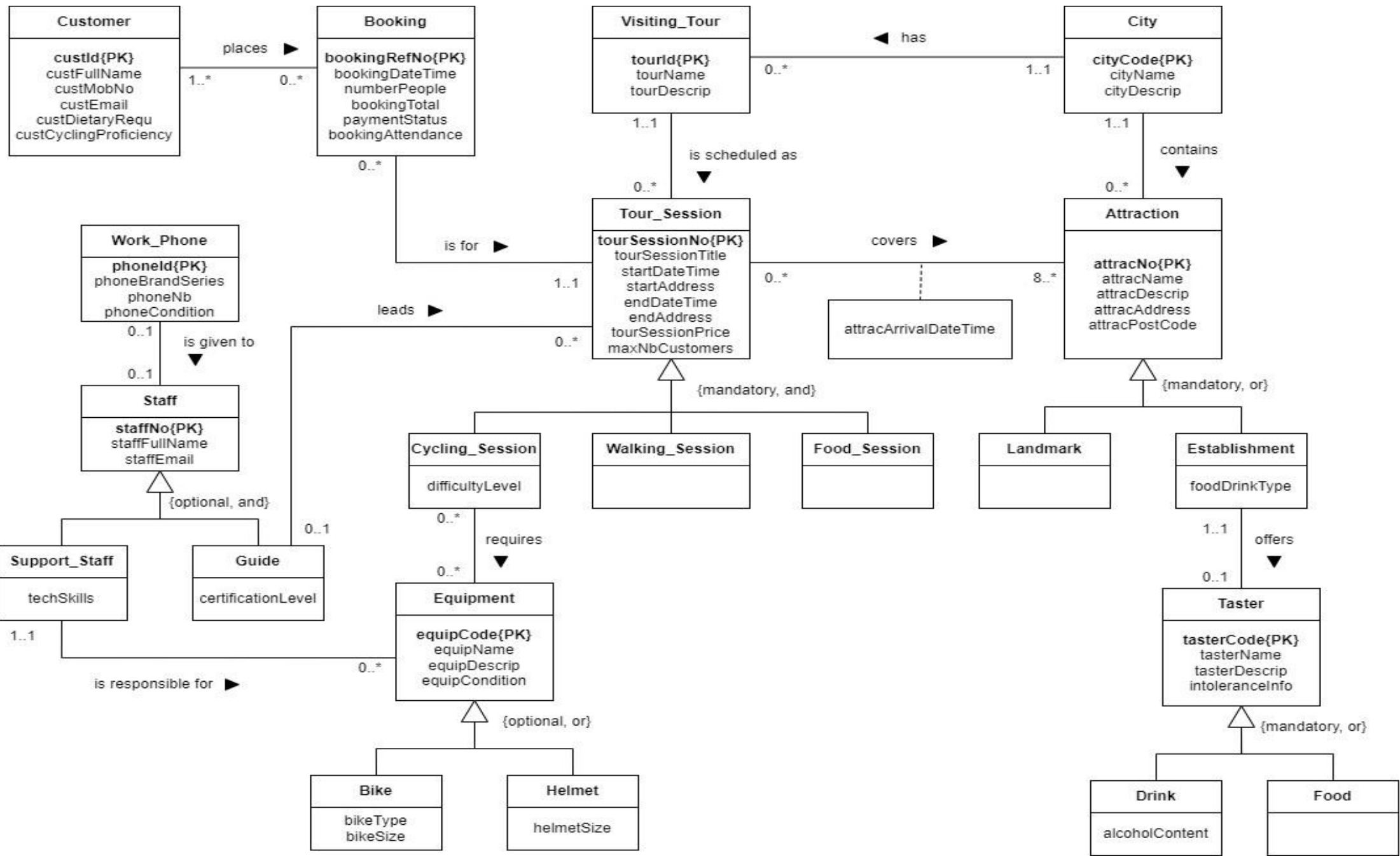


Figure 1. Conceptual EERD for Tourmato

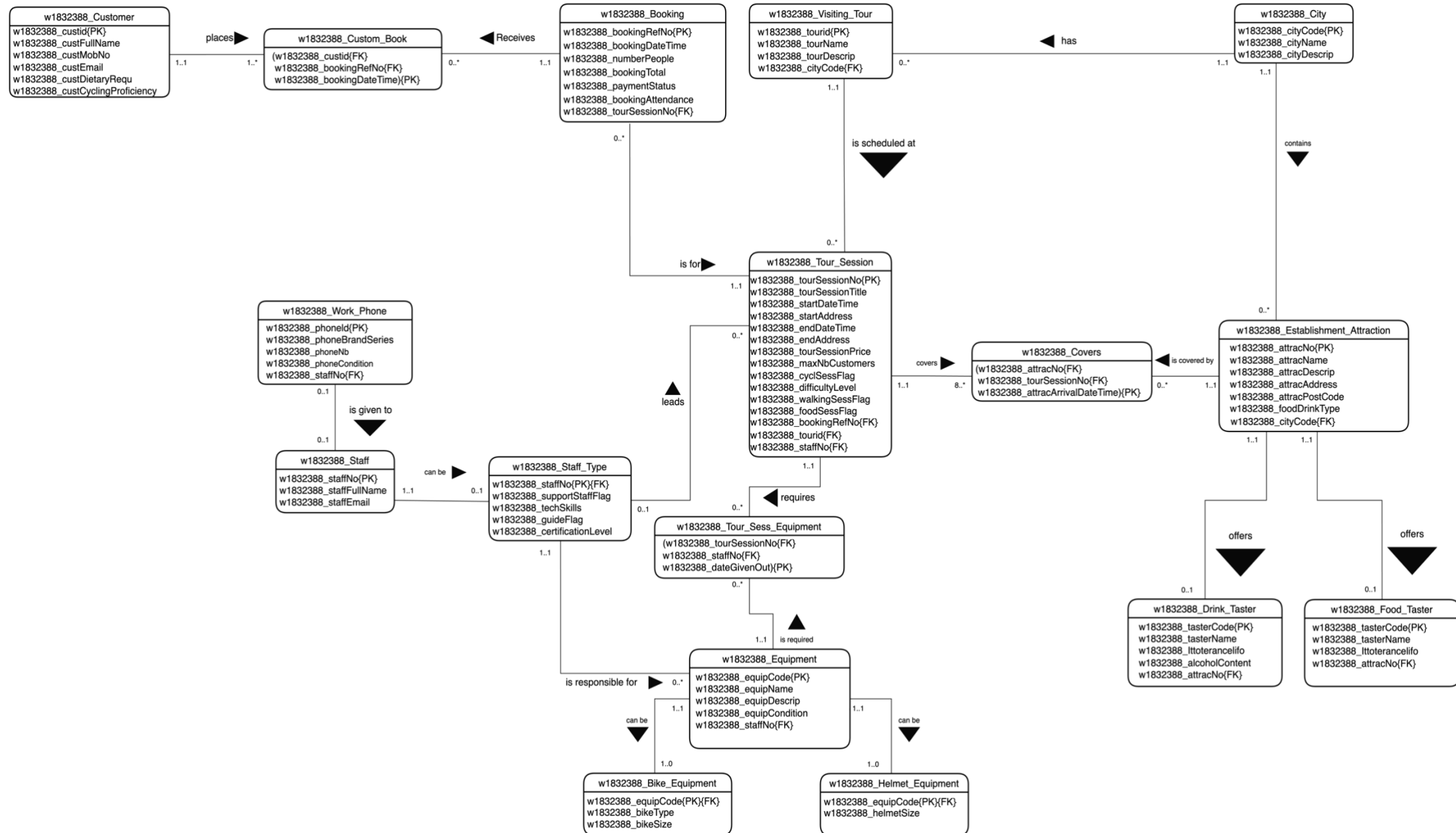


Figure 2: logical ERD of Tourmato Conceptual EERD

REPORT CONCLUDED